

Reasoning in LLMs



Yadollah Yaghoobzadeh

Reasoning

- ❑ The process of drawing **conclusions** based on **available information**.

Types of Reasoning

❑ Deductive

- is a type of reasoning in which a conclusion is drawn based on the truth of the premises.
- the conclusion must necessarily follow from the premises, meaning that if the premises are true, the conclusion must also be true.

For example:

- Premise: All mammals have kidneys.
- Premise: All whales are mammals.
- Conclusion: All whales have kidneys.

Types of Reasoning

□ Inductive reasoning

- is a type of reasoning in which a conclusion is drawn based on observations or evidence.
- the conclusion is likely to be true based on the available evidence, but it is not necessarily certain.

For example:

- Observation: Every time we see a creature with wings, it is a bird.
- Observation: We see a creature with wings.
- Conclusion: The creature is likely to be a bird.

Types of Reasoning

❑ Abductive reasoning.

- is a type of reasoning in which a conclusion is drawn based on the best explanation for a given set of observations.
- the conclusion is the most likely explanation based on the available evidence, but it is not necessarily certain.

For example:

- Observation: The car cannot start and there is a puddle of liquid under the engine.
- Conclusion: The most likely explanation is that the car has a leak in the radiator.

Types of Reasoning

- **Abductive:** Seeing a wet road and deducing it probably rained recently.
- **Inductive:** Observing that the road gets wet every time it rains, and concluding that rain causes the road to be wet.
- **Deductive:** ?

Reasoning Problems

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: The answer is **5**

Arithmetic Reasoning (AR)
(+ -×÷...)

Q: Take the last letters of the words in "Elon Musk" and concatenate them

A: The answer is **nk**.

Symbolic Reasoning (SR)

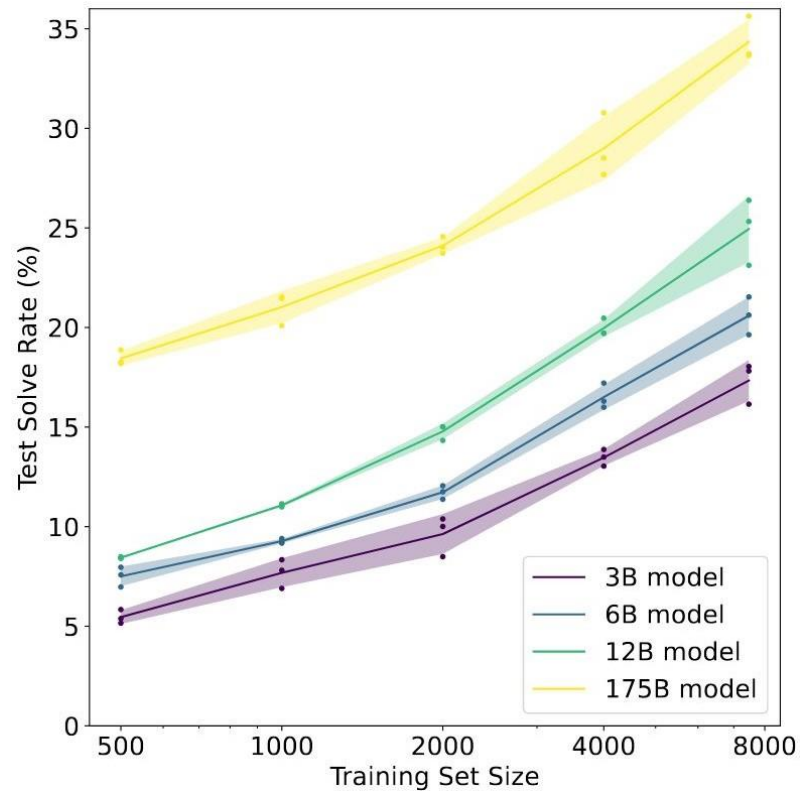
Q: What home entertainment equipment requires cable?
Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer is **(c)**.

Commonsense Reasoning (CR)

Reasoning Problems

Fine-tune GPT-3 on GSM8K (arithmetic): (Cobbe et al. 2021)



Conjecture: to achieve $> 80\%$, needs 100 times more fine-tuning data for 175B model

Reasoning Problems

Scaling up language model size does not **efficiently** achieve high performances, for Arithmetic Reasoning (*AR*), CommonSense Reasoning (*CR*) and Symbolic Reasoning (*SR*) tasks.

Reasoning Problems

Scaling up language model size does not **efficiently** achieve high performances, for Arithmetic Reasoning (*AR*), CommonSense Reasoning (*CR*) and Symbolic Reasoning (*SR*) tasks.

Proposed solution: **chain of thought prompting**

Chain of Thought Prompting

Chain of Thought (CoT)

Chain of Thought Prompting Elicits Reasoning in Large Language Models

Few-Shot CoT

Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma

Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou

Google Research, Brain Team
{jasonwei,dennyzhou}@google.com

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Large Language Models are Zero-Shot Reasoners

Zero-Shot CoT

Takeshi Kojima

The University of Tokyo
t.kojima@weblab.t.u-tokyo.ac.jp

Shixiang Shane Gu

Google Research, Brain Team

Machel Reid

The University of Tokyo

Yutaka Matsuo

The University of Tokyo

Yusuke Iwasawa

The University of Tokyo

Chain of Thought (CoT)

Definition:

A chain of thought is **a series of intermediate natural language reasoning steps** that lead to the final output.

Elements of a Prompt

- A prompt is composed with the following components:

- Instructions

- Context

- Input data

- Output indicator

Classify the text into neutral, negative or positive

Text: I think the food was okay.

Sentiment:

Chain of Thought (CoT)

(a) Few-shot

Examples

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. X

Chain of Thought (CoT)

(a) Few-shot

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A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT (Wei et al., 2022)

CoT Examples

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

Step-by-step Answer

Chain of Thought (CoT)

Examples

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

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Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(b) Few-shot-CoT (Wei et al., 2022)

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CoT Examples

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A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(d) Zero-shot-CoT (KoJima et al., 2022)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

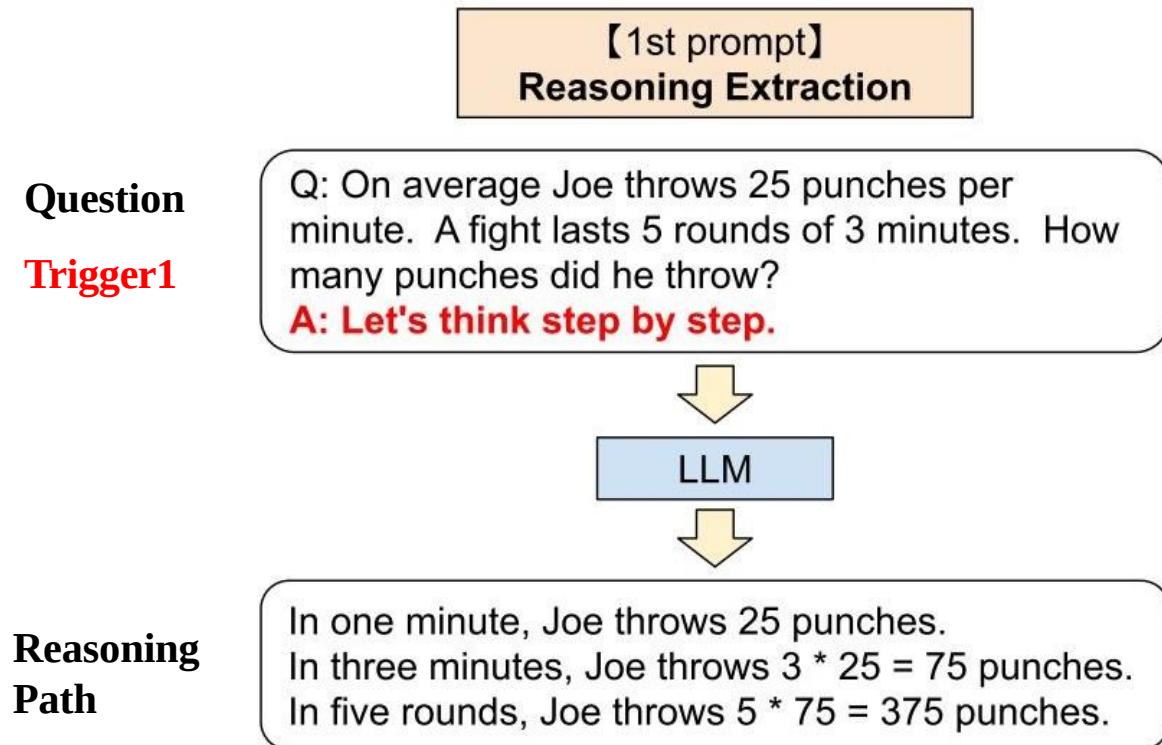
CoT Examples

Step-by-step Answer

Two-stage Prompting Step-by-step Answer

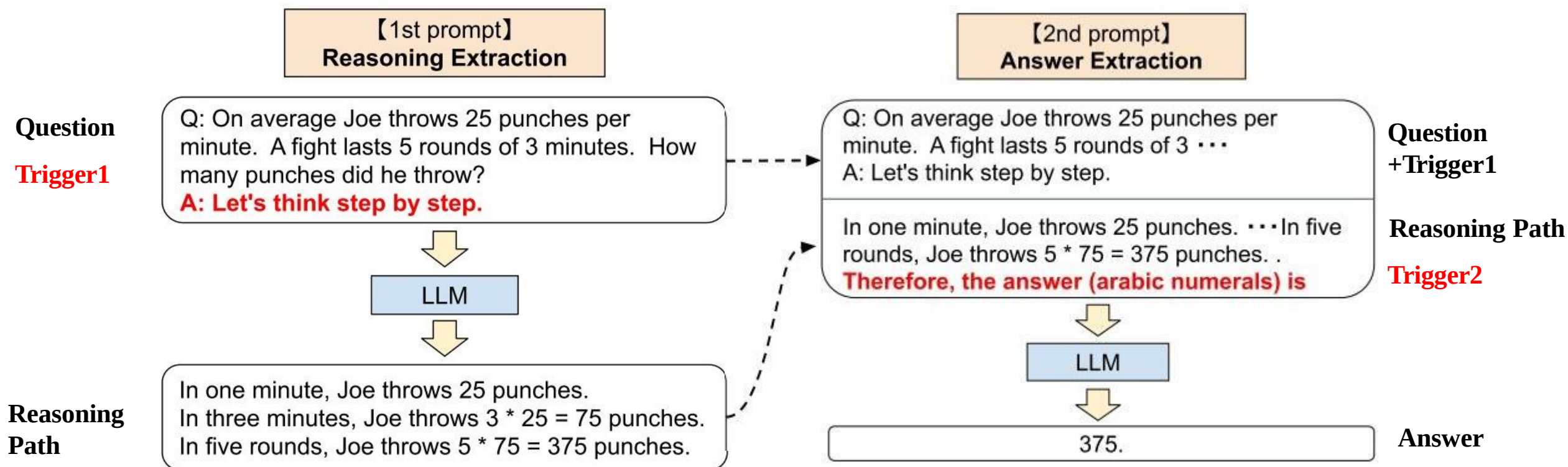
Zero-Shot Chain of Thought (CoT)

For **zero-shot CoT**, a **two-stage prompting** is applied:



Zero-Shot Chain of Thought (CoT)

For **zero-shot CoT**, a **two-stage prompting** is applied:



Experiments

Models

□ Pre-trained LLMs:

- Instruct GPT-3 (ada 350M, babbage 1.3B, curie 6.7B, and davinci 175B) (Ouyang et al., 2022)
- PaLM (8B, 62B, 540B) (Chowdhery et al., 2022)
- LaMDA (422M, 2B, 8B, 68B, 137B) (Thoppilan et al., 2022)
- GPT-3 (ada 350M, babbage 1.3B, curie 6.7B, davinci 175B)
- GPT-2 (1.5B)
- GPT-Neo (2.7B), GPT-J (6B), T0 (11B) (Sanh et al., 2022), OPT (13B) (Zhang et al., 2022)

Experiments

Arithmetic Reasoning

Free Response - Few-Shot CoT Prompt Exemplar

Free Response

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: **There are originally 3 cars. 2 more cars arrive. $3 + 2 = 5$.** The answer is **5**.

(Wei et al., 2022)

- **Manually** composed 8 exemplars

Free Response - Few-Shot CoT Prompt Exemplar

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(Wei et al., 2022)

Free Response

Q: Olivia has \$23. She bought five bagels for \$3 each. How much money does she have left?

A: **Olivia had 23 dollars. 5 bagels for 3 dollars each will be $5 \times 3 = 15$ dollars. So she has $23 - 15$ dollars left. $23 - 15$ is 8.** The answer is **8**.

(Wei et al., 2022)

You can have one or more equations.

Free Response - Few-Shot CoT Prompt Exemplar

Free Response

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

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Equations can be incomplete and combined math with words.

Free Response - Few-Shot CoT Prompt Exemplar

Free Response

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: **There are originally 3 cars. 2 more cars arrive. $3 + 2 = 5$.** The answer is **5**.

(Wei et al., 2022)

- **Manually** composed **8** exemplars
- All contains equations with flexible formats
- Benchmarked on:
 - **GSM8K** (Cobbe et al. 2021)
 - **SVAMP** (Patel et al., 2021)
 - **MAWPS** (Koncel-Kedziorski et al., 2016)

Multiple Choice - Few-Shot CoT Prompt Exemplar

Multiple Choice

Q: A person is traveling at 20 km/hr and reached his destiny in 2.5 hr then find the distance?

Answer Choices: (a) 53 km (b) 55 km (c) 52 km (d) 60 km (e) 50 km

A: **The distance that the person traveled would have been 20 km/hr * 2.5 hrs = 50 km. The answer is (e).**

Multiple Choice

Q: If $a / b = 3/4$ and $8a + 5b = 22$, then find the value of a.

Answer Choices: (a) $1/2$ (b) $3/2$ (c) $5/2$ (d) $4/2$ (e) $7/2$

A: **If $a / b = 3/4$, then $b = 4a / 3$. So $8a + 5(4a / 3) = 22$. This simplifies to $8a + 20a / 3 = 22$, which means $44a / 3 = 22$. So a is equal to $3/2$. The answer is (b).**

The exemplars have **various** formats.

Multiple Choice - Few-Shot CoT Prompt Exemplar

Multiple Choice

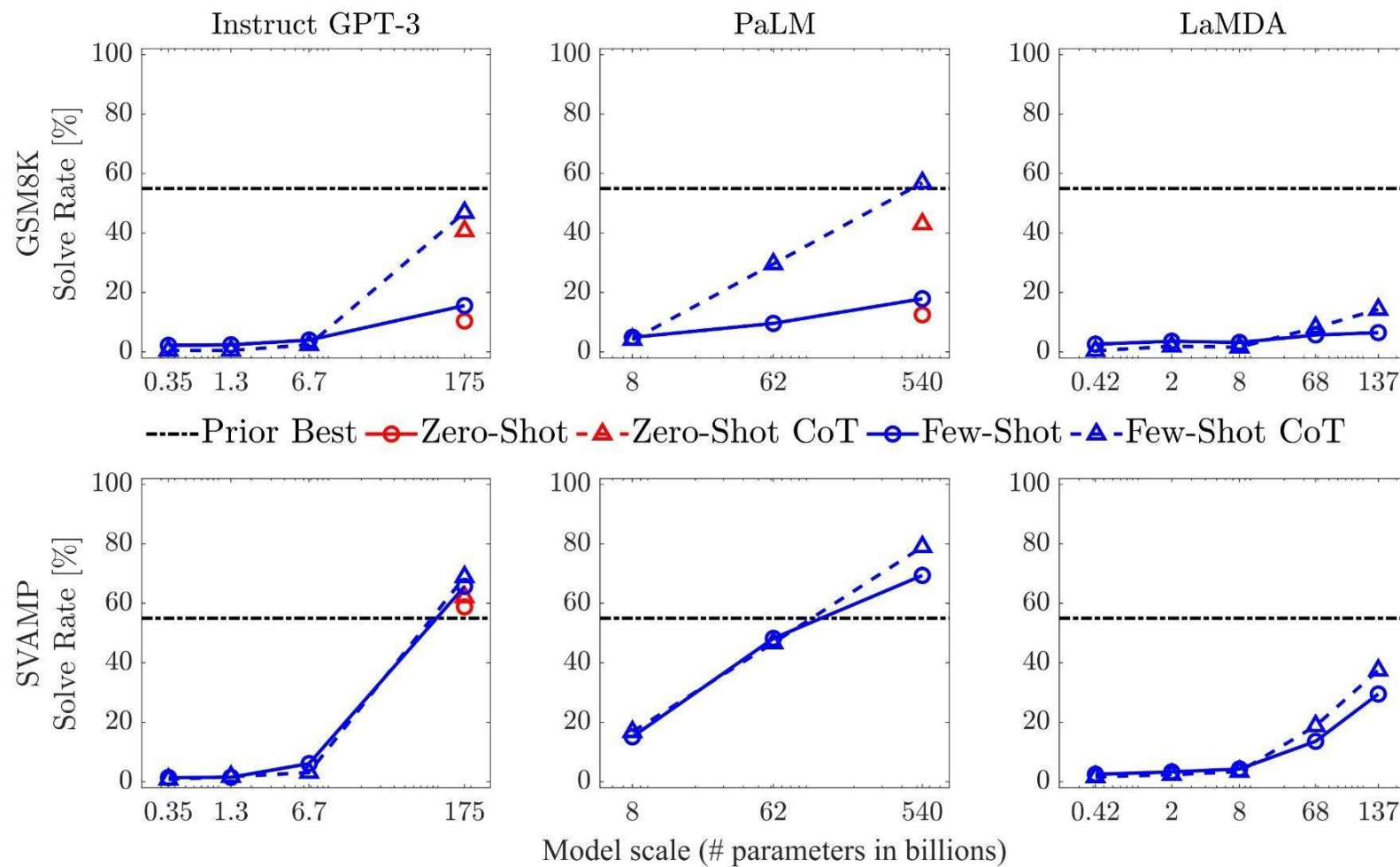
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A: **The distance that the person traveled would have been 20 km/hr * 2.5 hrs = 50 km.** The answer is **(e)**.

- 4 exemplars, whose questions, intermediate reasoning, and answers are from **training set**
- Exemplars have flexible formats
- Benchmarked on **AQuA-RAT (Ling et al., 2017)**

Arithmetic Reasoning - Results



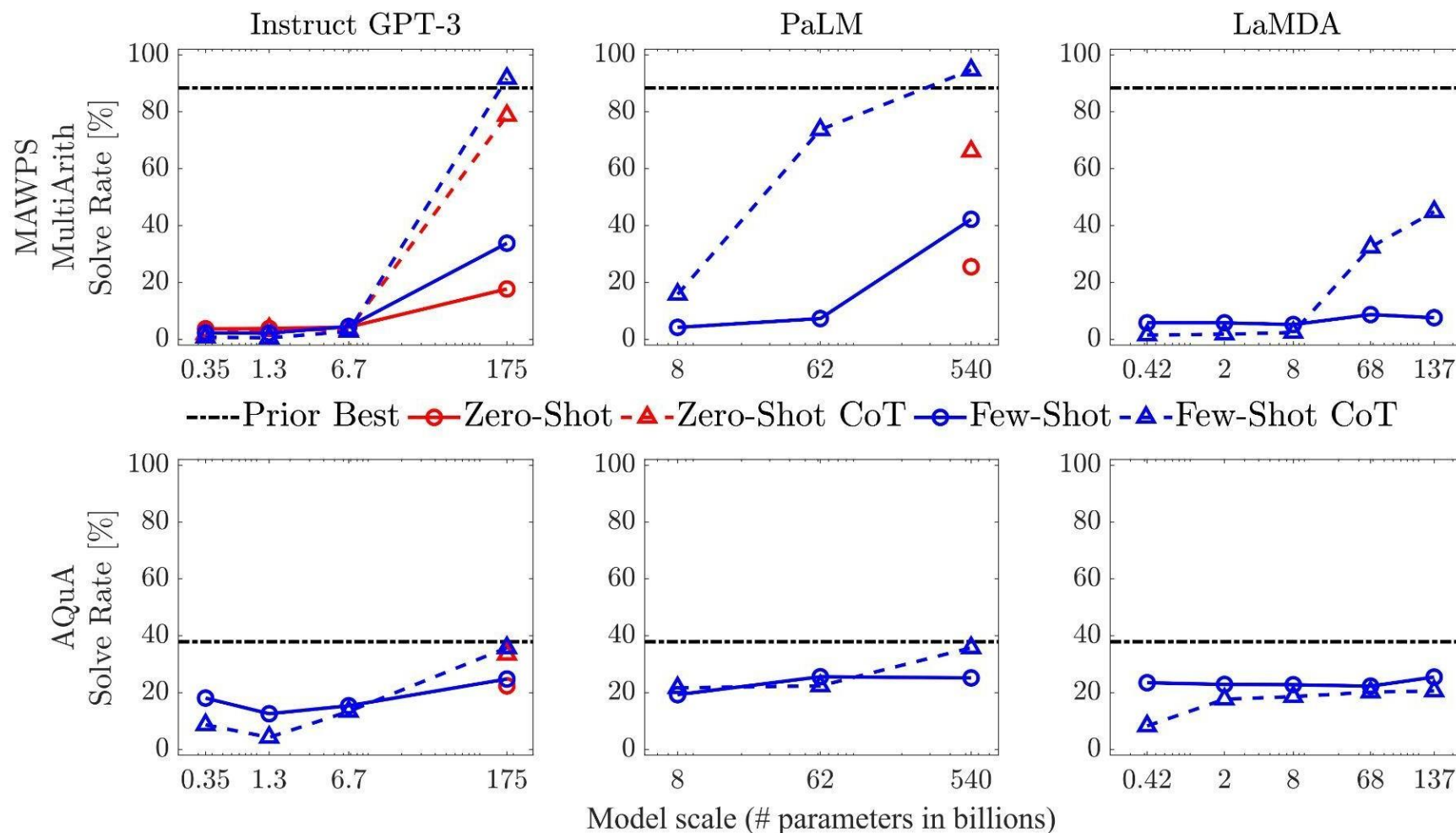
GSM8K

Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

SVAMP

Each pack of dvds costs 76 dollars. If there is a discount of 25 dollars on each pack. How much do you have to pay to buy each pack?

Arithmetic Reasoning - Results



MAWPS - MultiArith

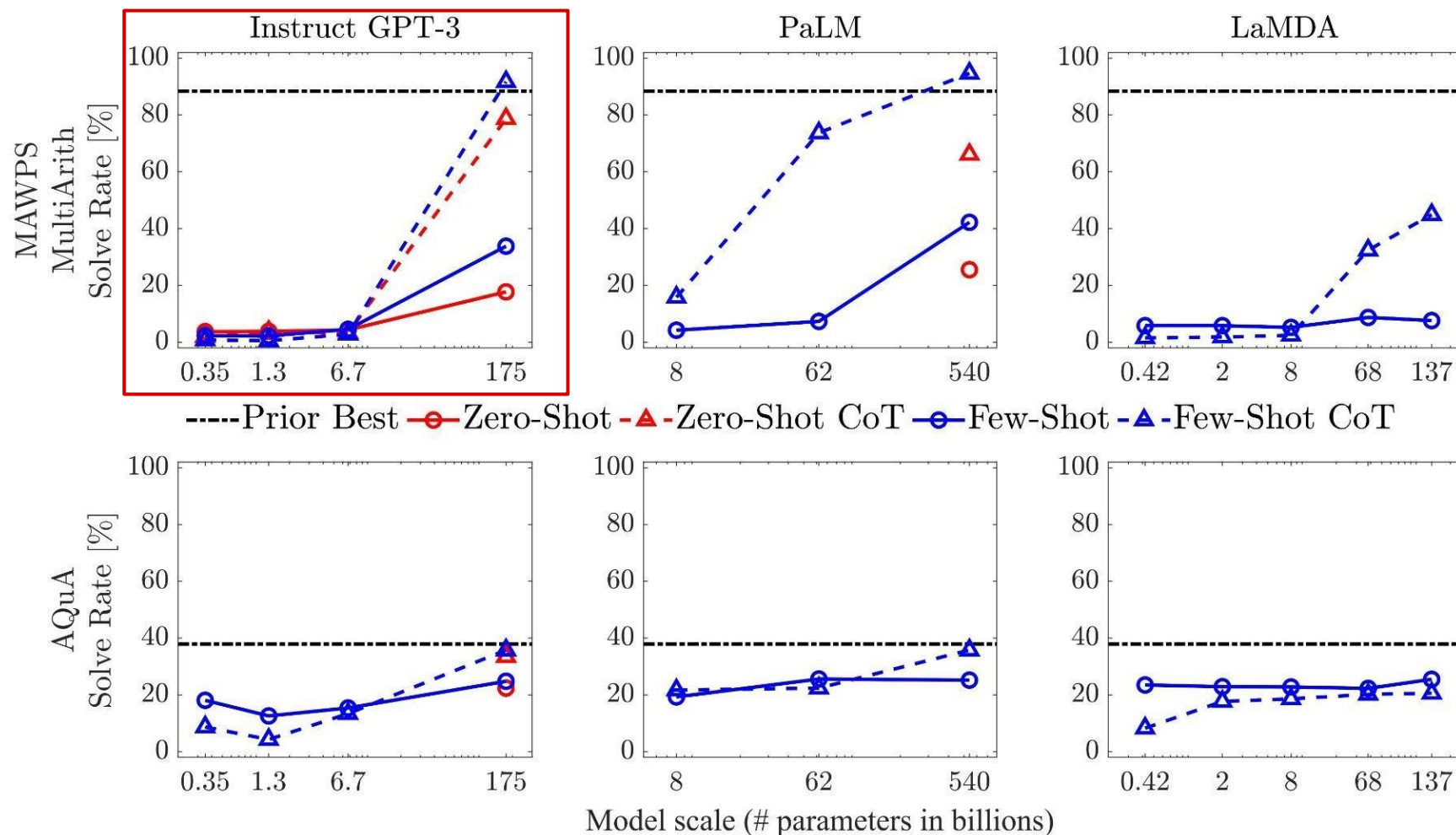
The school cafeteria ordered 42 red apples and 7 green apples for students lunches. But, if only 9 students wanted fruit, how many extra did the cafeteria end up with?

AQuA-RAT

A person is traveling at 20 km/hr and reached his destiny in 2.5 hr then find the distance?

Answer Choices: (a) 53 km (b) 55 km (c) 52 km (d) 60 km (e) 50 km

Arithmetic Reasoning - Results



MAWPS - MultiArith

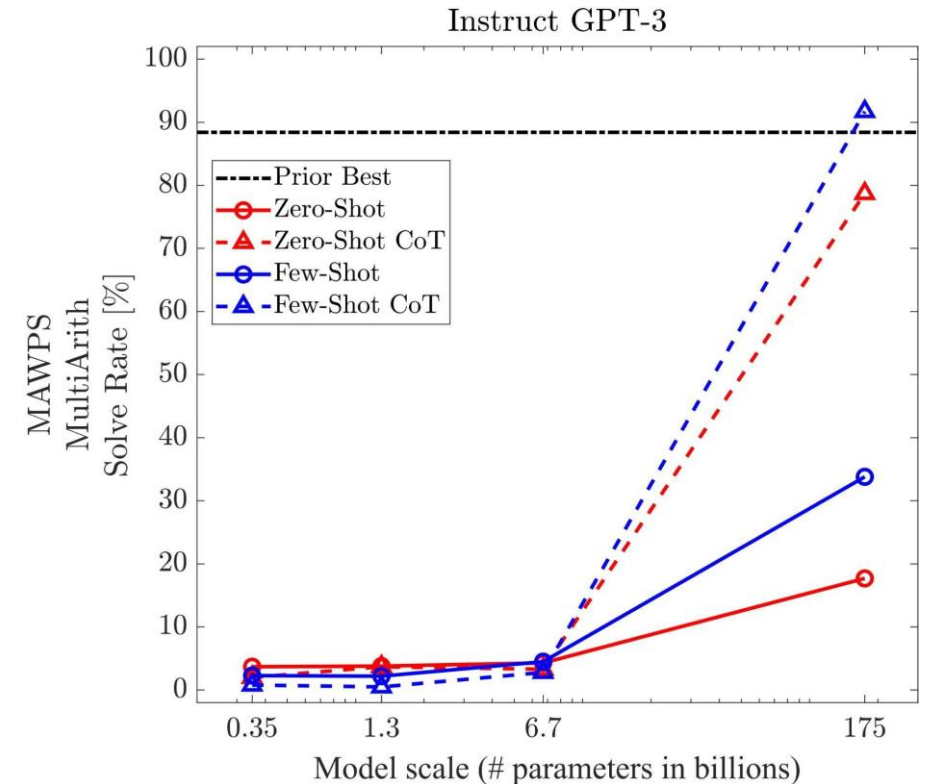
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Arithmetic Reasoning - Observations

- Both zero-shot and few-shot chain of thought promptings are emergent **abilities of model scale**.
- They do not positively impact performance for small models, but start to yield performance gains when used with models with more than ~100B parameters.
- Few-shot CoT achieves **better** performance on LLM than zero-shot CoT.
- Instruct GPT-3: text-davinci-002 achieves similar performance as PaLM 540B model



Experiments

Symbolic Reasoning

Symbolic Reasoning - Last Letter Concatenation

Last letter concatenation

Q: Take the last letters of the words in "Elon Musk" and concatenate them

A: The last letter of "Elon" is "n".
The last letter of "Musk" is "k".
Concatenating them is "nk". The answer is **nk**.

- Generate full names by **randomly concatenating** names from the **top one-thousand first and last** names from name census data
- 4 exemplars with **strict** format

Symbolic Reasoning - Coin Flip

Coin Flip

Q: A coin is heads up. Tom does not flip the coin. Mike does not flip the coin. Is the coin still heads up?

A: The coin was flipped by no one. So the coin was flipped 0 times. The coin started heads up, and it was not flipped, so it is still heads up. So the answer is yes.

Coin Flip

Q: A coin is heads up. Jamey flips the coin. Teresa flips the coin. Is the coin still heads up?

A: The coin was flipped by Jamey and Teresa. So the coin was flipped 2 times, which is an even number. The coin started heads up, so after an even number of flips, it will still be heads up. So the answer is yes.

8 exemplars with **strict** format.

Symbolic Reasoning - In & Out-of-domain Test

Last letter concatenation

Q: Take the last letters of the words in "Elon Musk" and concatenate them

A: The last letter of "Elon" is "n".
The last letter of "Musk" is "k".
Concatenating them is "nk". The answer is **nk**.

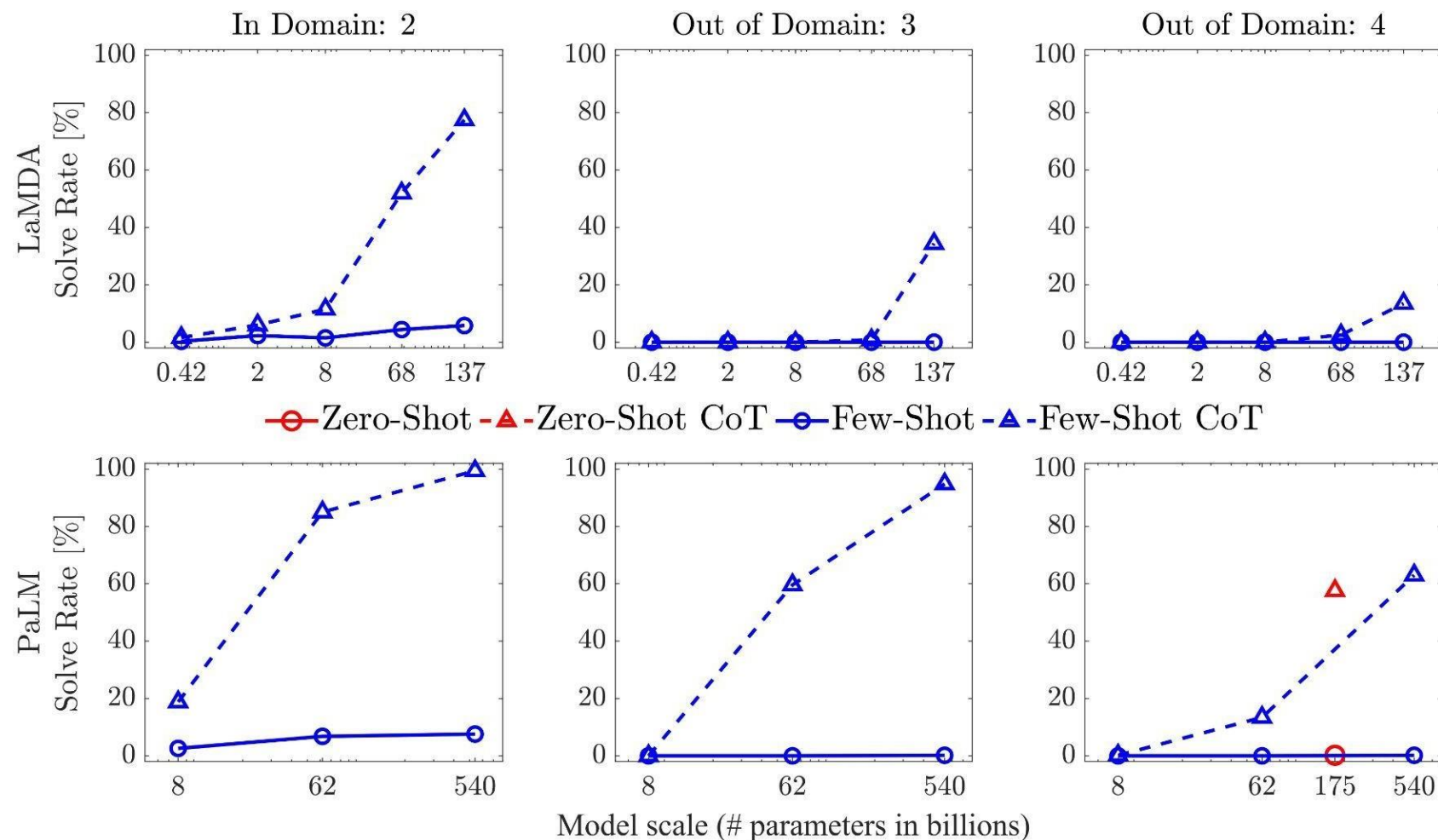
Coin Flip

Q: A coin is heads up. Tom does not flip the coin. Mike does not flip the coin. Is the coin still heads up?

A: The coin was flipped by no one. So the coin was flipped 0 times. The coin started heads up, and it was not flipped, so it is still heads up. So the answer is **yes**.

- In-domain test set: examples had the same number of steps as the few-shot exemplars
- Out-of-domain (OOD) test set: examples had more steps than those in the exemplars.

Symbolic Reasoning - Last Letter Concatenation



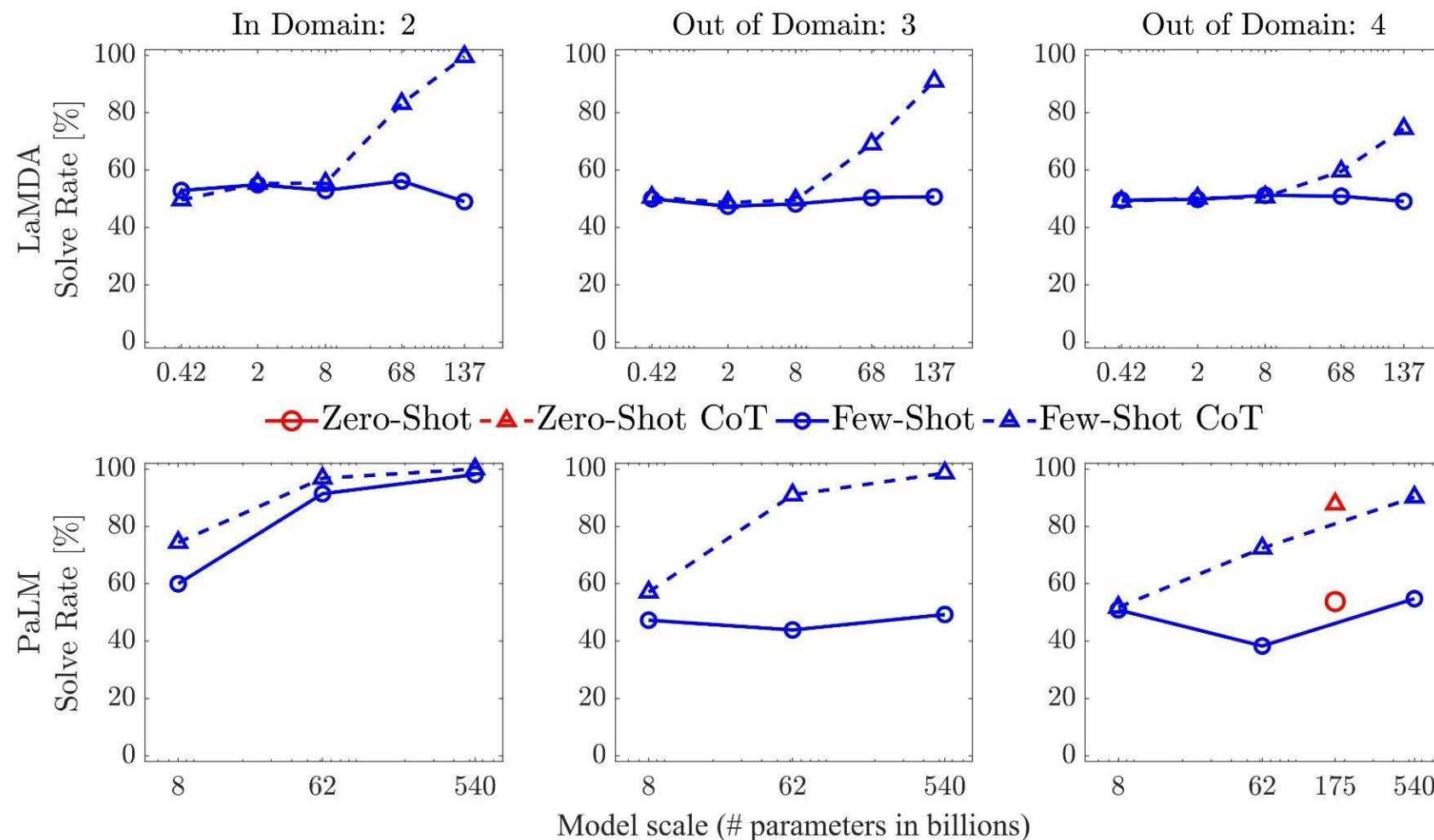
In-Domain

Take the last letters of the words in "**Elon Musk**" and concatenate them.

Out-of-Domain

Take the last letters of the words in "**Johann Sebastian Bach**" and concatenate them.

Symbolic Reasoning - Coin Flip



In-Domain

A coin is heads up. **Tom does not flip the coin. Mike does not flip the coin.** Is the coin still heads up?

Out-of-Domain

A coin is heads up. **Tom does not flip the coin. Mike does not flip the coin. Jake flips the coin.** Is the coin still heads up?

Symbolic Reasoning - Observations

- Standard prompting **fails out-of-domain** tests for both tasks.
- Few-shot CoT prompting is **emergent abilities of model scale**.
- CoT does not positively impact performance for small models, but start to yield performance gains when using models with more than ~100B parameters for both in-domain and out-of-domain tests.
- **Zero-shot** CoT using **Instruct-GPT-3 175B** achieves the similar performance as **few-shot** CoT in both tasks using **540B PaLM model**.

Commonsense Reasoning - Toy Problems

CSQA (Talmor et al., 2019)

Q: What home entertainment equipment requires cable?

Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer is **(c)**.

StrategyQA (Geva et al., 2021)

Q: Did Aristotle use a laptop?

A: The answer is **no**.

Sport Understanding

Q: Is the following sentence plausible? "Jamel Murray was perfect from the line."

A: The answer is **yes**.

Date Understanding

Q: 2015 is coming in 36 hours. What is the date one week from today in MM/DD/YYYY

A: So the answer is **01/05/2015**.

Commonsense Reasoning - Toy Problems

SayCan Robot Planning

Locations = [counter, table, user, trash, bowl].

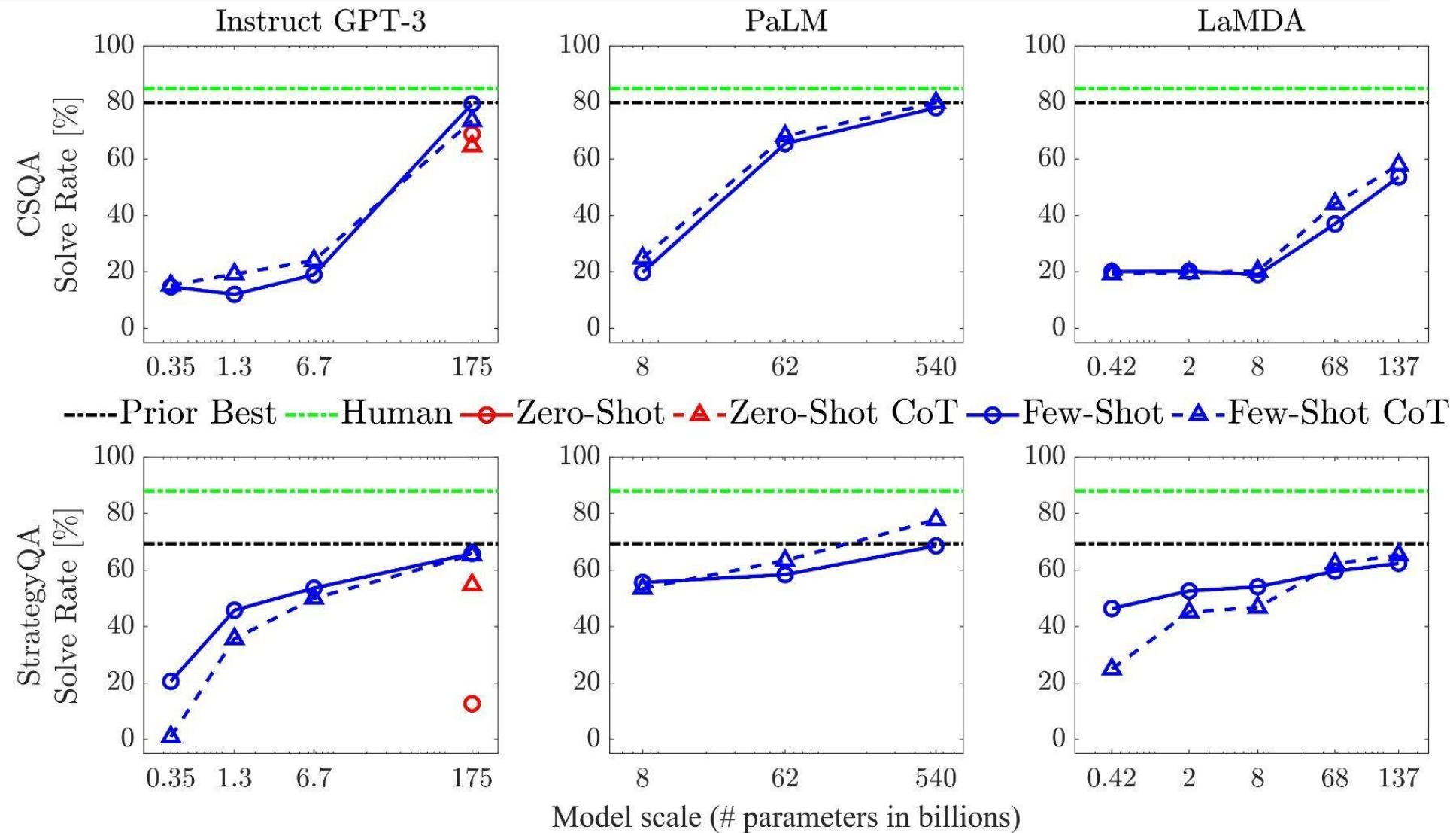
Objects = [7up, apple, kettle chips, tea, multigrain chips, coke, lime soda, jalapeno chips, rice chips, orange, grapefruit soda, pepsi, redbull, energy bar, sponge, water].

The robot can pick up items with **pick(object)** and put down items with **put(object)** as well as find objects or locations with **find()**. The robot can only understand the explicit locations and objects listed.

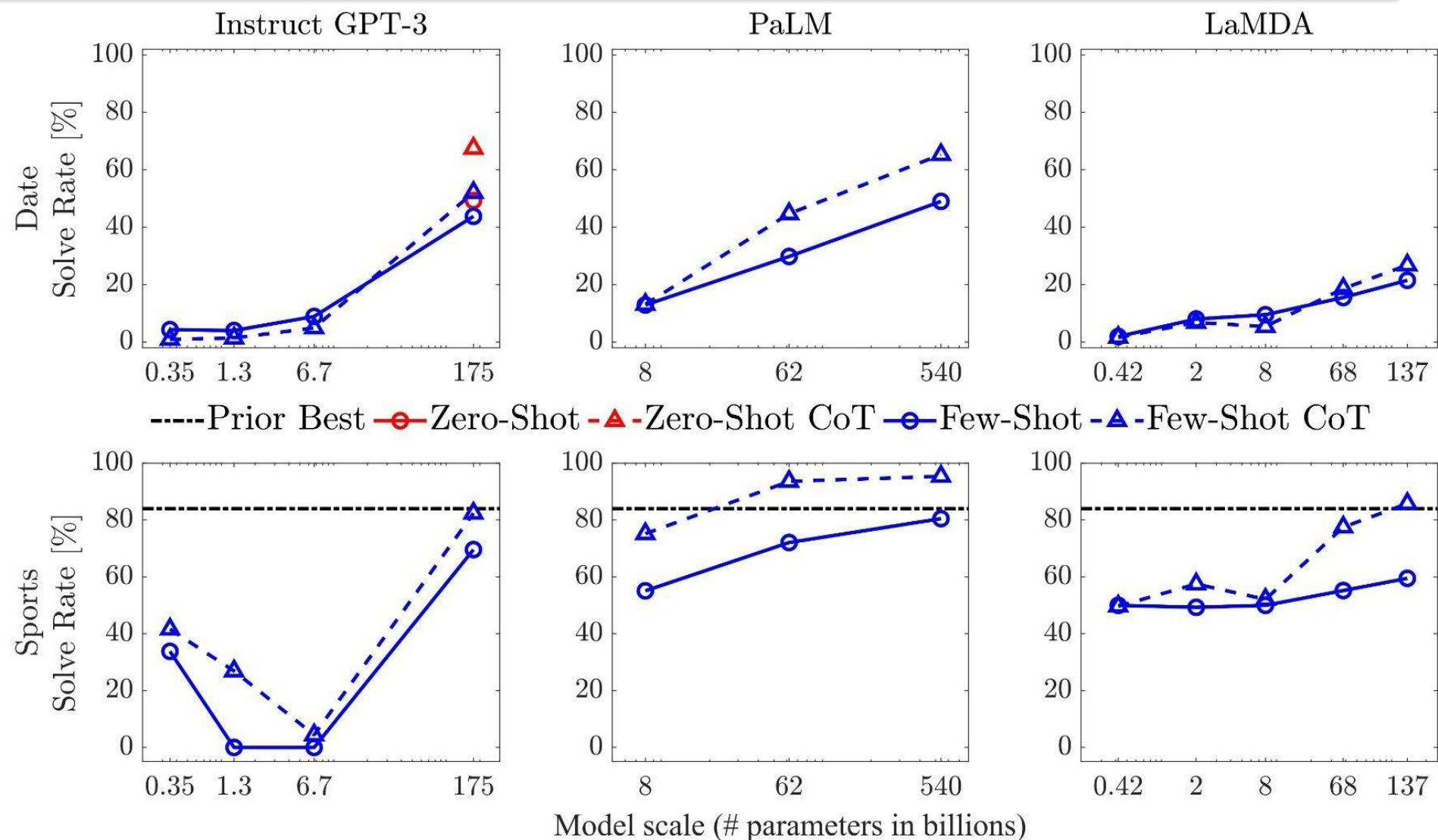
Human: How would you throw away a redbull?

Plan: 1. **find(redbull)**, 2. **pick(redbull)**, 3. **find(trash)**, 4. **put(redbull)**, 5. **done()**.

Commonsense Reasoning - Results



Commonsense Reasoning - Results

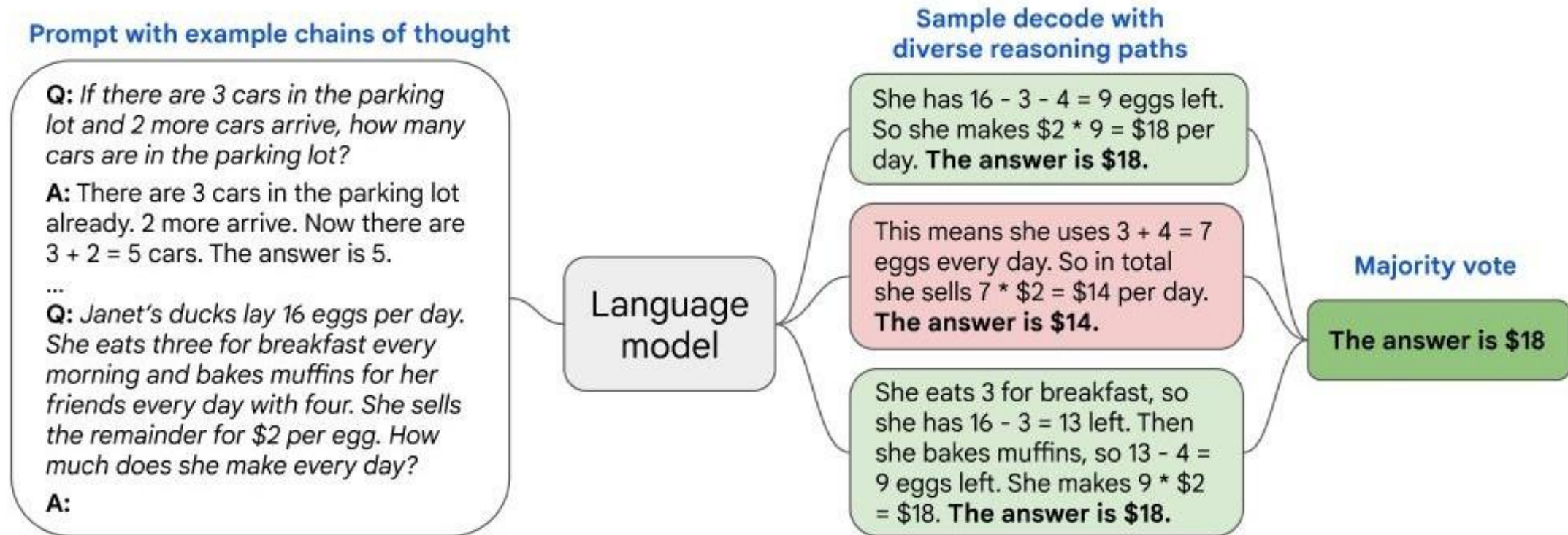


Commonsense Reasoning - Observations

- For all tasks, scaling up **model size** improved the performance of standard prompting.
- CoT prompting led to further gains, with improvements appearing to be **largest for PaLM 540B**.
- CoT has **minimal** benefits on CSQA and StrategyQA tasks.
- **Few-shot** achieves better performance than **Zero-shot** CoT on 175B GPT-3 model for CSQA and Strategy QA tasks, but **Zero-shot** CoT shows significant improvement for **Date understanding** task.

More Advances - Self-Consistency

Change greedy decode (single-path) to self-consistency (multi-path) in few-shot CoT:



Wang, Xuezhi, et al. "Self-consistency improves chain of thought reasoning in language models." arXiv preprint arXiv:2203.11171 (2022).

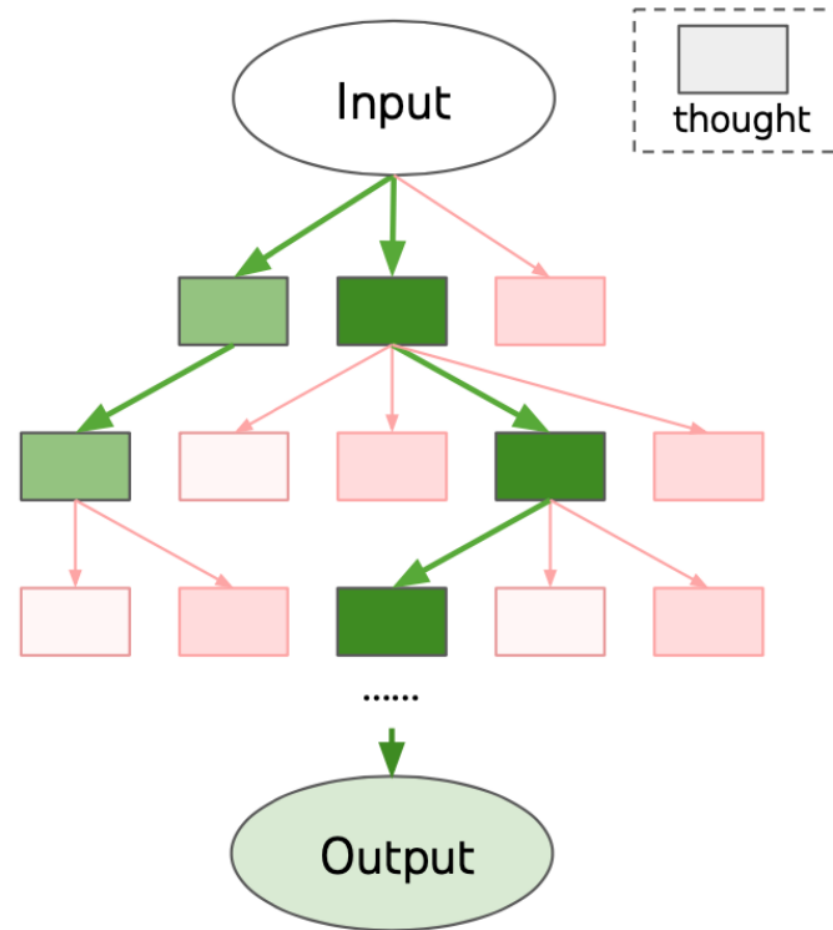
More Advances - Self-Consistency

Showcase results on **AR**, **CR** tasks:

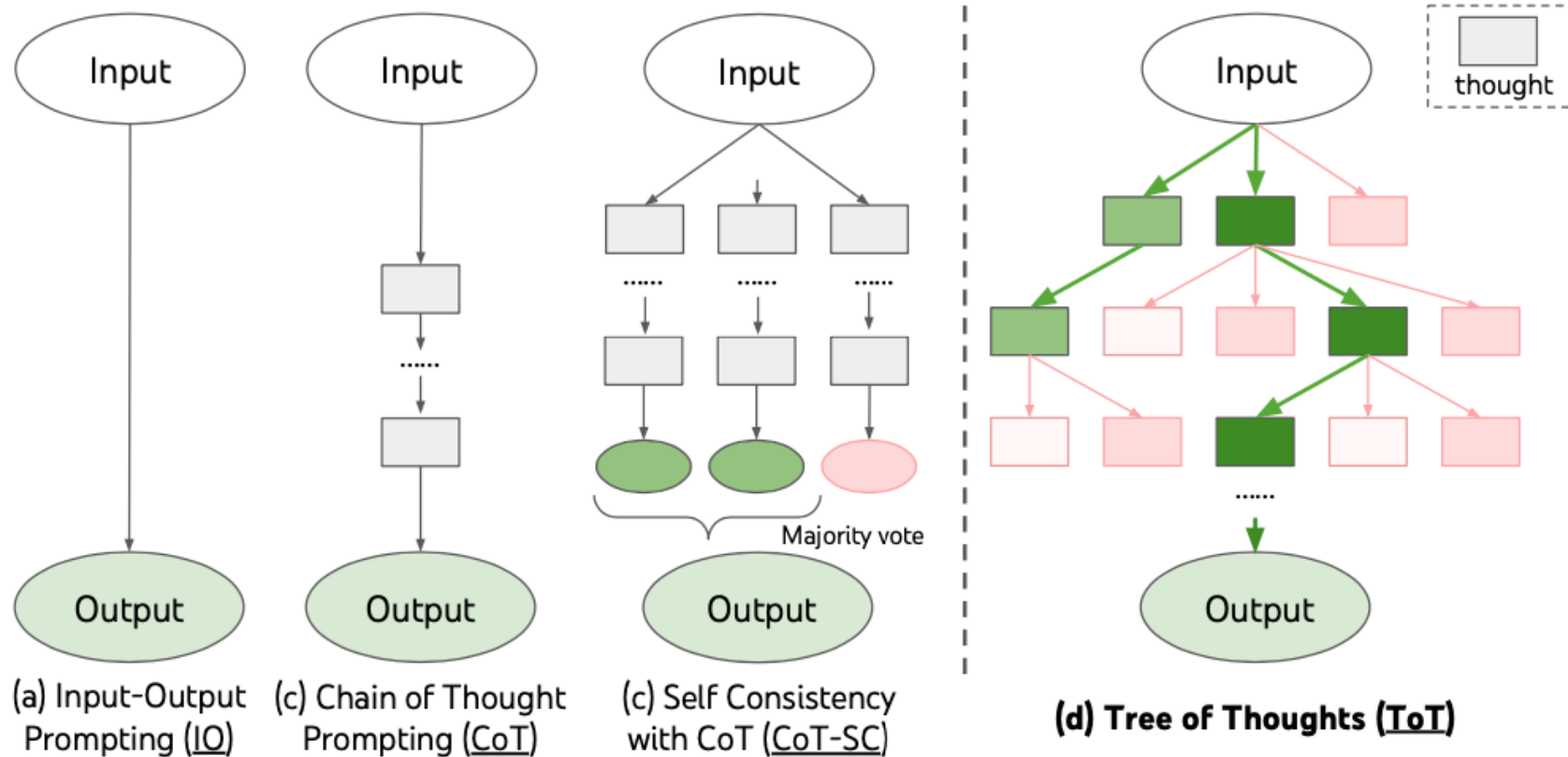
	Method	GSM8K	CommonsenseQA
	Previous SoTA	35 ^e / 57 ^g	91.2^a
LaMDA (137B)	Greedy decode (Single-path)	17.1	57.9
	Self-Consistency (Multi-path)	27.7 (+10.6)	63.1 (+5.2)
PaLM (540B)	Greedy decode (Single-path)	56.5	79.0
	Self-Consistency (Multi-path)	74.4 (+17.9)	80.7 (+1.7)

Wang, Xuezhi, et al. "**Self-consistency improves chain of thought reasoning in language models.**" arXiv preprint arXiv:2203.11171 (2022).

Tree of Thoughts (ToT)



ToT vs IO, CoT, CoT-SC



Different survey papers

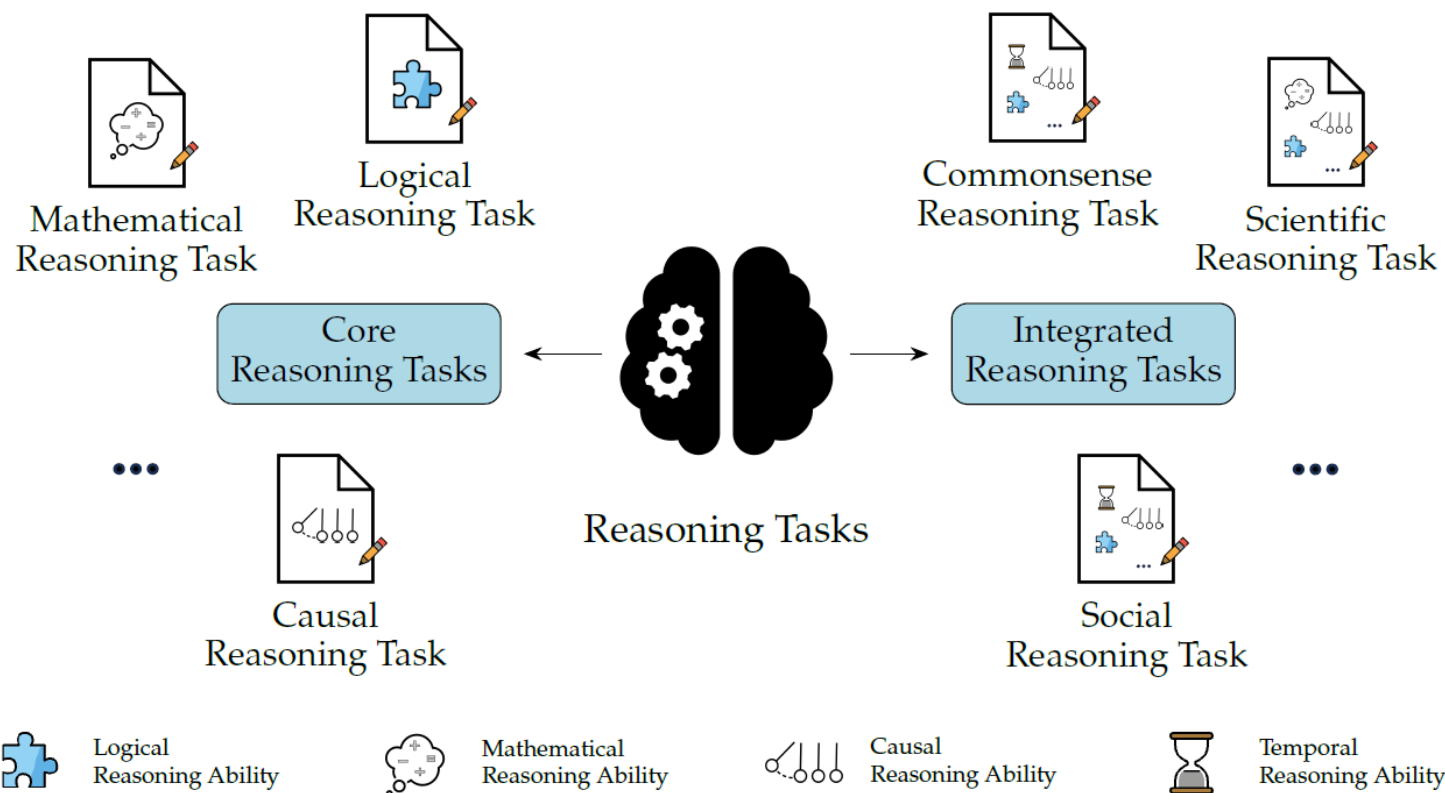


Figure 1: Schematic overview of the two types of reasoning tasks distinguished in this survey. *Core* reasoning tasks are designed to assess a particular reasoning ability within a given context. Conversely, *integrated* reasoning tasks involve the concurrent use of various reasoning skills. Tasks and abilities listed are not exhaustive.